Cardiovascular System (CARD) · NPLEX-P1-CARD-17

CV-17 Cardiovascular Pathogens

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-17-01-C01				
NPLEX-P1-CARD-17-01-C02				
NPLEX-P1-CARD-17-01-C03				
NPLEX-P1-CARD-17-01-C04				
NPLEX-P1-CARD-17-01-C05				
NPLEX-P1-CARD-17-01-C06				
NPLEX-P1-CARD-17-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C01-BD1

Rebuild the four endocarditis cocci

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C01 Activity: NPLEX-P1-CARD-17-01-C01-BD1 Date: _____

Without looking at your notes, write out the four classic gram-positive cocci of infective endocarditis. For each one write its arrangement on Gram stain, whether it is catalase positive or negative, what the coagulase or hemolysis or growth test shows, and the named virulence factors it uses. Then write one sentence per organism saying how those factors let it attach to a valve and how fast it destroys the leaflet. Finish by writing the three bench results that separate staphylococci from streptococci and enterococci, and the one test that separates Staphylococcus aureus from Staphylococcus epidermidis.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C01-BD2

Two patients, two valves, two time courses

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C01 Activity: NPLEX-P1-CARD-17-01-C01-BD2 Date: _____

Patient one injects drugs and becomes ill over three days with a loud tricuspid murmur and a perforated leaflet found at surgery. Patient two had a dental cleaning six weeks ago, has a known bicuspid aortic valve, and has felt tired and feverish ever since. For each patient write the likely organism, the portal of entry, why that organism can or cannot colonize the valve as it was before the infection, and the virulence factors that explain the speed of leaflet damage. Then write why protein A and bound coagulase make neutrophils and antibody less effective, and write what a sterile platelet-fibrin deposit on a scarred valve does for an organism that invades poorly. Finish by writing which of the two patients you would expect to have an abscess and why.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-17-01-C01-CP01

The four gram-positive cocci of infective endocarditis

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Staphylococcus aureus	Viridans streptococci	Enterococcus faecalis	Staphylococcus epidermidis
Arrangement on Gram stain				
Catalase result				
Other bench tests that identify it				
Named virulence factors				
How it attaches to the valve surface				
Condition of the valve it needs				
Usual portal of entry				
Speed of valve destruction and course				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C02-BD1

Why blood cultures stay negative

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C02 Activity: NPLEX-P1-CARD-17-01-C02-BD1 Date: _____

Write the definition of culture-negative endocarditis and then list its causes in order, starting with the commonest. For each cause write the organism type, the exposure or body site it comes from, and the reason routine blood cultures fail to grow it. Spell out what the letters of HACEK stand for and write where those organisms normally live. For Coxiella burnetii write the cell compartment it multiplies in, the form that survives drying, the material people inhale, and the test used instead of culture. For the two Bartonella species write the vector or animal source for each. Finish by writing the bench profile of Streptococcus gallolyticus and the one test that separates it from Enterococcus.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C02-BD2

Match the exposure to the organism and read the bench results

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C02 Activity: NPLEX-P1-CARD-17-01-C02-BD2 Date: _____

Four adults have endocarditis with no growth in routine blood cultures. One works on a goat farm and helps at kidding. One lives outdoors and has body lice. One has poor dental health and large friable vegetations that have embolized. One took an antibiotic for a sore throat the week before cultures were drawn. For each person write the likely cause, the structural or metabolic feature of the organism that explains why it did not grow, and the kind of test that would identify it. Then a different patient grows catalase-negative cocci in chains that are nonhemolytic, hydrolyze esculin in bile, and do not grow in 6.5% NaCl. Write the organism, the body site it came from, and what its presence in the blood suggests about that site. Finish by writing what the result would have been instead if the isolate had grown in 6.5% NaCl.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-17-01-C02-Q01

A group D organism from the bench tests

A 71-year-old man comes to the physician because of 5 weeks of fatigue, low-grade fever, and weight loss. Temperature is 38.1 °C (100.6 °F). A new murmur is heard at the cardiac apex, and echocardiography shows a vegetation on the mitral valve. Blood cultures grow gram-positive cocci in chains. The isolate is catalase negative and nonhemolytic, hydrolyzes esculin in the presence of bile, and fails to grow in 6.5% sodium chloride. Which of the following organisms is most consistent with these findings?

A. Enterococcus faecalis

B. Streptococcus mitis

C. Streptococcus gallolyticus

D. Staphylococcus epidermidis

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C03-BD1

The two human forms and every route of transmission

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C03 Activity: NPLEX-P1-CARD-17-01-C03-BD1 Date: _____

Write the name of the organism that causes Chagas disease and the kind of organism it is. Describe its two human forms side by side: for each one write whether it is inside or outside cells, whether it has an external flagellum, and whether it divides. Write where in the body the dividing form multiplies and name the four cell types it favors. Write the insect vector, where the vector lives, when it feeds, and the part of the insect that carries the infective parasite, then write why the bite itself does not transmit it. List every other route of transmission named in the notes. Finish by writing what the acute phase looks like, how long it lasts, what is seen on a stained blood smear during it, and the two local signs of entry with the site each one follows.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C03-BD2

Predict chronic Chagas disease from the cells the parasite lives in

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C03 Activity: NPLEX-P1-CARD-17-01-C03-BD2 Date: _____

A woman who grew up in rural Bolivia and moved away 30 years ago has had no symptoms since. Write the name of the phase she has been in, why serology rather than a blood smear is used to find infection in that phase, and the approximate share of infected people who go on to chronic disease. Then, working only from the cell types the parasite infects, predict what three decades of parasite persistence and inflammation do to the heart: write what replaces the lost myocytes, what happens to chamber size and contractility, where a thin-walled aneurysm forms and what can collect in it, and which conduction pattern is typical. Predict what the loss of enteric plexus neurons does to the esophagus and colon, and explain in two sentences why those organs dilate. Finish by writing why a symptom-free carrier living far from any triatomine bug still matters to a blood bank and a transplant program.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-17-01-C03-RT01

Rebuild the Trypanosoma cruzi life cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C03 Activity: NPLEX-P1-CARD-17-01-C03-RT01 Date: _____

Rebuild the eight-step life cycle of Trypanosoma cruzi from memory, one step per line and in order, starting with the bug taking a blood meal. Beside each step write which parasite form is present at that point. Draw a line across the page to separate the steps that happen in the insect from the steps that happen in the human. Then circle the two steps where the parasite multiplies, and write one sentence saying why the number of parasites in human blood can rise even though the circulating form does not divide.

Rebuild from memory: Life cycle of Trypanosoma cruzi

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C04-BD1

Two tick-borne bacteria from the ground up

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C04 Activity: NPLEX-P1-CARD-17-01-C04-BD1 Date: _____

Draw Borrelia burgdorferi and Rickettsia rickettsii side by side and label the structural features of each. For Borrelia write where its flagella sit, what their rotation does for the cell, what its outer membrane is rich in and what it lacks, and how its genome is arranged. For Rickettsia write its shape and Gram character and the two metabolic reasons it must live inside a host cell. For both write the stains used to see them and why Gram stain is not useful. Then write the tick genus that carries each one, the reservoir for each, and which tick stage transmits most human Lyme disease. Finish by writing the region of the United States where most Rocky Mountain spotted fever cases occur.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C04-BD2

Explain the block and explain the rash

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C04 Activity: NPLEX-P1-CARD-17-01-C04-BD2 Date: _____

Patient one was camping in Connecticut, has an expanding red skin lesion, and now has a heart rate of 44 per minute with a varying degree of atrioventricular block that changes within hours. Patient two was in Oklahoma, has had fever for 4 days, and now has macules on the wrists and ankles spreading to the trunk, palms, and soles, with falling blood pressure and falling platelets. For each patient name the organism, the tick that carried it, and the tissue the organism is damaging, then explain the cardiovascular finding from the way that organism moves and survives. For patient one write where in the conduction system the block usually sits and why it can change degree hour by hour. For patient two write how the organism gets from one endothelial cell to the next and explain how damaged endothelium produces edema, hypotension, and a petechial rash. Finish by writing why a tick removed within the first day rarely transmits Lyme disease, naming the surface proteins involved.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-17-01-C04-RT01

Rebuild the Borrelia surface switch from tick to host

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C04 Activity: NPLEX-P1-CARD-17-01-C04-RT01 Date: _____

Rebuild the seven-step sequence that takes Borrelia burgdorferi from an unfed tick gut into human skin, one step per line and in order. Beside the first step write which surface protein anchors the spirochete in the gut and what it binds. Beside the step where the proteins change, write which protein falls and which rises, and what triggers the change. Beside the salivary gland step write the tick protein that the new surface protein binds and what that does for the spirochete. Write the number of hours of tick attachment the whole sequence usually needs. Then write the last step separately: name the surface lipoprotein that keeps changing in the host, write the genetic mechanism that changes it, and write in one sentence what that does to antibody made against the earlier version.

Rebuild from memory: Borrelia surface switching from tick to host

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C05-BD1

Flaviviruses and filoviruses side by side

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C05 Activity: NPLEX-P1-CARD-17-01-C05-BD1 Date: _____

Write the two virus families that cause viral hemorrhagic fever in this chapter and name the example viruses in each. For each family write the shape of the virion, whether it is enveloped, the genome type and strand sense, the reservoir, the route of transmission, and the first cells it infects. Then write what a positive-sense genome does on its own inside a cell and what a negative-sense virion must carry with it, and write which of the two naked genomes would be infectious if injected alone. For yellow fever write the organ and the zone of the liver lobule that dies, the name of the dying cell on the slide, and the two consequences of lost hepatocyte function. For dengue write the number of serotypes, the cells it infects, when in the illness the plasma leak peaks, and the secreted viral protein that damages the endothelial glycocalyx.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C05-BD2

Why the second dengue infection is the dangerous one

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C05 Activity: NPLEX-P1-CARD-17-01-C05-BD2 Date: _____

A man had dengue 4 years ago and recovered. He now has a second infection with a different serotype and is much sicker than the first time. Write why immunity from the first serotype does not protect him, then rebuild the mechanism in order: what the old antibody does to the new virus, which receptor on which cells takes the coated virus in, what that route of entry does to viral load, and what it does to the interferon response. Write what cross-reactive memory T cells add. Then explain why an infant with waning maternal antibody can have severe disease at a first infection. Separately, write what a rising hematocrit with falling platelets tells you about where the plasma has gone, and why the patient gets worse as the fever breaks. Finish by writing how the mechanism of bleeding in Ebola differs from this one, naming the two interferon-blocking proteins and the macrophage product that starts disseminated intravascular coagulation.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-17-01-C05-PR01

Predict the course of a second dengue infection

A traveler who had mild DENV-1 infection 4 years ago is now on day 5 of a febrile illness caused by DENV-2, and her fever is falling. Predict each item below, write the direction of the change, and give the mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What the DENV-1 antibody does when it meets DENV-2		
What happens to the number of virus particles entering monocytes and macrophages		
What happens to the type I interferon response in those cells		
What happens to the amount of fluid in the vessels and to the hematocrit		
What happens to the platelet count		
What happens to her blood pressure and alertness as the fever falls		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C06-BD1

Coxsackie B from gut to dilated heart

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C06 Activity: NPLEX-P1-CARD-17-01-C06-BD1 Date: _____

Write the definition of myocarditis and the commonest identified cause of it in high-income countries. For coxsackie B virus write the family and genus, the genome type and strand sense, the shape of the capsid, and whether it has an envelope, then write the three things that missing envelope lets the virus survive and the route of spread that follows from it. Rebuild the seven-step sequence that runs from entry at the mouth to a dilated ventricle, one step per line and in order. Beside the receptor step write both surface molecules the virus uses, where in the heart muscle one of them concentrates, and why that matters in a newborn. Beside the protease step write the two host proteins protease 2A cuts and what each cut does to the cell. Finish by writing the other two syndromes this virus causes and what often happens to the pericardium alongside the myocardium.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C06-BD2

Explain the findings and judge a biopsy result

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C06 Activity: NPLEX-P1-CARD-17-01-C06-BD2 Date: _____

A 14-year-old had diarrhea in August and now has chest pain, a rub, a troponin of 3.2 ng/mL (reference under 0.04 ng/mL), and frequent ectopic beats with ST and T wave changes. Explain each finding from the mechanism of injury: write why troponin appears in the blood, why the chest pain is pleuritic with a rub, why the rhythm is unstable, and why contractility falls. Write the structural protein whose loss lets the membrane tear under contractile stress, name the inherited disease with the same structural failure, and explain why the two diseases damage the same link in different ways. Then write what happens over the following weeks when T cells arrive, and why that phase can leave a dilated ventricle even after the virus is cleared. Finish with this result: a heart biopsy from a different patient is positive for parvovirus B19 DNA. Write why that result does not by itself establish the cause, and write what parvovirus B19 binds and which vessels it injures.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-17-01-C06-Q02

The torn sarcolemma in coxsackie myocarditis

A 9-year-old boy is brought to the physician in September because of 4 days of fever, pleuritic chest pain, and shortness of breath that began a week after an episode of watery diarrhea. Temperature is 38.4 °C (101.1 °F) and a pericardial friction rub is heard. Serum troponin is elevated. A nonenveloped, positive-sense single-stranded RNA virus is recovered from a stool specimen. Which of the following best explains the tearing of the myocyte sarcolemma in this illness?

A. Viral protease cleavage of dystrophin, which links sarcomeric actin to the sarcolemma

B. ADP-ribosylation of elongation factor 2 and arrest of myocyte protein synthesis

C. Polymerization of host actin into tails that push the organism between cells

D. Nonneutralizing antibody carrying virus into cells through Fc gamma receptors

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-17-01-C07-BD1

Diphtheria toxin and the tuberculous pericardium

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C07 Activity: NPLEX-P1-CARD-17-01-C07-BD1 Date: _____

For Corynebacterium diphtheriae write the shape of the rod, how the cells lie beside each other, the granules inside them, and the colony color on tellurite agar. Write which strains can make toxin and what they carry, then write the one nutrient condition that turns toxin production on and the repressor that holds it off. Rebuild the toxin mechanism in order, from the B portion binding its receptor to the molecule that stops working, and write what the cell does next. Write what forms in the throat and what it is made of, then write when myocarditis appears in the illness, what the muscle looks like microscopically, and which conduction problems follow. For Mycobacterium tuberculosis write the staining property and the wall component behind it, the route by which it usually reaches the pericardium, the kind of granuloma and effusion it produces, and what the two pericardial layers become as the inflammation heals.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-17-01-C07-BD2

Predict the physiology of a fixed pericardium and a starved aortic media

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C07 Activity: NPLEX-P1-CARD-17-01-C07-BD2 Date: _____

A man has a thick calcified pericardial shell after treated tuberculosis. Working from the fact that the shell fixes the total volume of the heart, predict what happens to early diastolic filling and what stops it, the extra heart sound this produces, the shape of the jugular venous pulse, and what happens to jugular venous pressure during inspiration and why. Write what the diastolic pressures in the four chambers do relative to each other, and list the findings that follow from systemic venous congestion. Then a second man has untreated syphilis of 25 years. Write which vessels the treponemes persist around, what those vessels supply, and what the narrowing of those vessels does to the outer media. Predict which part of the aorta fails and why that part rather than another, the appearance of the intima over the scarred media, and the three consequences of a dilated aortic root and arch. Finish by writing why atherosclerotic aneurysms instead favor the infrarenal abdominal aorta.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-17-01-C07-RT01

Rebuild the path from vasa vasorum to aneurysm

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-17 Cardiovascular Pathogens Competency: NPLEX-P1-CARD-17-01-C07 Activity: NPLEX-P1-CARD-17-01-C07-RT01 Date: _____

Rebuild the seven-step sequence that runs from treponemes in the adventitia to an aortic aneurysm, one step per line and in order. Beside the step where the small vessels narrow, write the name of that lesion and the cells that cuff the vessels. Beside the step where the media is affected, write which two structural components of the media are lost and what the wall can no longer do. Then draw a cross section of the thoracic aorta and label the adventitia, the vasa vasorum, the outer media, and the intima, and mark the layer that becomes ischemic. Finish by writing one sentence saying which does the damage, the organism itself or the loss of blood supply to the wall.

Rebuild from memory: From vasa vasorum endarteritis to aortic aneurysm

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-17-01-C07-CA01

Chest discomfort and a widened ascending aorta in a man with a remote untreated infection

A 58-year-old man comes to the physician because of 6 months of chest discomfort on exertion and increasing breathlessness when lying flat. He recalls a painless genital ulcer and a rash on his palms and soles in his early thirties, neither of which was treated. Temperature is 37.1 °C (98.8 °F), pulse is 88/min and collapsing in character, respirations are 18/min, and blood pressure is 160/48 mm Hg. A high-pitched early diastolic murmur is heard at the left sternal border. Chest radiography shows a widened ascending aorta and arch with a thin rim of calcification. At the time of later surgery the ascending aorta is dilated, the intima is wrinkled into a coarse tree-bark pattern, and the aortic valve ring is stretched with structurally normal leaflets.

1. Recognize. Name the organism, the stage of its infection, and the usual interval between untreated infection and this presentation, and name the lesion that produced the tree-bark intima.

2. Explain. Explain how the organism damages the aortic wall, starting with where it persists and ending with the loss of elastic fibers and smooth muscle, and explain why the valve leaflets are normal while the valve leaks.

3. Differentiate. Write how this aneurysm differs from an atherosclerotic aortic aneurysm in location and in mechanism, and name the feature of the aortic wall that explains the difference in site.

4. Investigate. Write which microscopy methods show this organism and why Gram stain does not, and write what you would look for in the aortic wall and in the adventitial vessels on histology.

5. Decide. State which structural change produces the exertional chest discomfort, and name the two other structures a large aneurysm of the arch can compress.

6. Justify. Justify the mechanism from the blood supply of the thoracic aortic media, explaining why ischemia of the outer media rather than direct destruction by the organism is what weakens the wall.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____